EV Market Insights

Project Report · Mahi Patidar · Nov 2, 2026

1. Introduction

1.1 Project Overview

EV Market Insights turns two electric-vehicle datasets into a project website, an interactive dashboard page and two Tableau Public views: an executive dashboard and a market story. The website presents a Flask web application, the Tableau dashboard and the Tableau story as one project.

The larger dataset covers 200 manufacturers, 17,980 models, 15 vehicle categories, 7 market regions, 10 manufacturing countries and model years 2018 to 2026. Its records add up to 1.55 billion annual sales units. Each record has 36 attributes, including battery capacity, range, charging speed, price, safety rating and customer rating. The dashboard page uses a second, 100,000-row file with the same columns.

The GitHub repository MahiPatidar8/EV-Market- holds both datasets, the project website and the interactive dashboard page. The live Tableau dashboard and story are published on Tableau Public and linked from the website.

1.2 Purpose

The project exists to let a reader answer EV market questions in seconds instead of reading raw tables. Its goals are:

- Present the EV market in a clear website with an overview, features, analytics and resources.
- Let users filter the dashboard page by year, region, category, battery type, drivetrain, use case, manufacturing country and manufacturer.
- Let each chart switch between 12 measures, such as price, range, rating, resale value and sales.
- Share the analysis as links, with a Tableau dashboard and story that open in any web browser.

The source tables appear to be generated rather than real registration data, because totals are almost identical across regions and years. The findings therefore demonstrate the analysis method and are not market facts.

2. Ideation Phase

2.1 Problem Statement

A 200,000-row EV dataset with 36 attributes is too large to read by hand, and a Tableau workbook can only be opened by people who have Tableau. Analysts, students and planners need a fast way to see which regions, vehicle categories, battery types and manufacturers drive the EV market.

How can a large EV dataset be turned into a filterable, shareable view that answers market questions in seconds?

2.2 Empathy Map Canvas

The canvas below describes the main user, an analyst or student who explores the EV market to prepare a report or a decision.

Quadrant	What the user shows
Says	"Which region sells the most EVs?" "Does battery type change range or rating?"
Thinks	The raw data is too big to scan. I need totals and comparisons, not 200,000 rows.
Does	Opens the dashboard, filters by region or battery type, compares categories and manufacturers, copies the summary table.
Feels	Frustrated by static tables and by tools that need installation. Confident when every chart updates together.

2.3 Brainstorming

Four ways to present the data were considered, and the repository delivers all of them around one Tableau analysis.

Idea	Decision
Keep the analysis only in Tableau workbooks	Kept as the core analysis, published on Tableau Public so it opens in a browser
Dashboard page with nine filters, 12 measures and a summary table	Built as the EV Market Insights dashboard page

Idea	Decision
Guided market story with the main findings	Built as a Tableau Public story
One project website that links every piece	Built as the EV Market Insights project showcase

Features chosen: nine filters that update every chart at once, a measure selector on each chart, six KPI tiles that show how many of the 100,000 vehicles match, and a sortable summary table that can be copied.

3. Requirement Analysis

3.1 Customer Journey Map

A user moves from opening the project website to sharing a filtered view in five steps.

Stage	User action	Touchpoint	Outcome
Open	Visits the project website	Hero, overview and the three views	Understands what the project covers
Explore	Opens the dashboard page	Six KPI tiles, four measure charts, price vs range chart	Sees market size, price, range and rating at once
Filter	Picks years, region, category, battery type or manufacturer	Nine filters and a status line	Narrows the market to one slice
Understand	Opens the Tableau dashboard and story	Tableau Public views linked from the website	Gets the main findings in a guided form
Share	Sends a link or copies the summary table	Website links and the Copy table button	Colleagues see the same views

3.2 Solution Requirements

The solution must meet the functional and non-functional requirements below.

Functional

- Show six KPIs: vehicle models, average price, average range, average rating, total annual sales and three-year resale value.

- Show charts by region, battery type, vehicle category and model year, each with a choice of 12 measures.
- Show price against range on a log price scale, coloured by drivetrain, from a random sample of up to 3,000 matching vehicles.
- Filter every chart together by year range, region, category, battery type, drivetrain, use case, manufacturing country and manufacturer.
- Provide a sortable summary table grouped by eight dimensions, with a copy button and a show-all-rows option.
- Show a clear message when no vehicle matches the filters.
- Publish a Tableau executive dashboard and a market story, and link both from the project website.

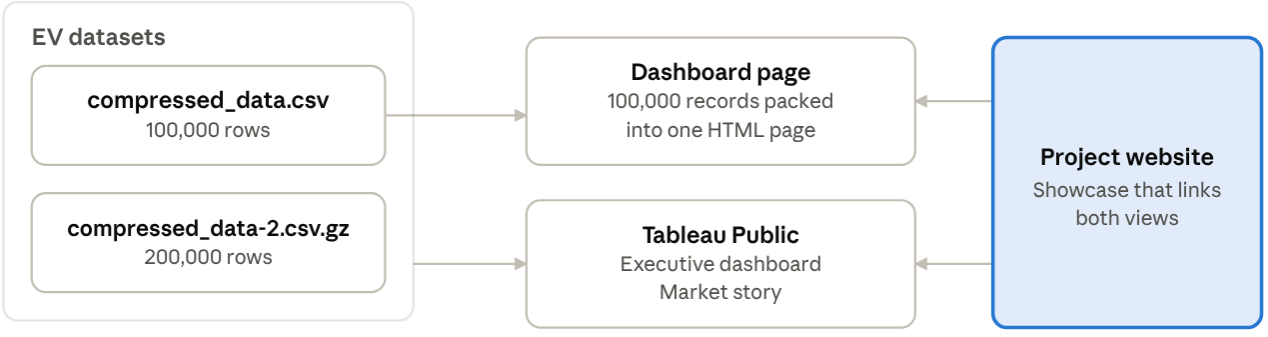
Non-functional

- Redraw every chart on each filter change from data already loaded in the page.
- Embed the dashboard data in the page, so no data server is needed.
- Follow the viewer's light or dark colour scheme and the reduced-motion setting.
- Open in any modern browser, on desktop or phone.
- Run the website and dashboard page as static files with no server.

3.3 Data Flow Diagram

Data moves from two compressed datasets into the dashboard page and the Tableau views, and the website links them together.

One website links the views built on the EV datasets



data flow · 2 datasets, 2 views, 1 website

The dashboard page embeds the 100,000-row file, and the website links to it and to the Tableau Public dashboard and story.

3.4 Technology Stack

Layer	Technology	Role
Analysis	Tableau, published on Tableau Public	Executive dashboard and market story
Data	CSV and gzip-compressed CSV, 100,000 and 200,000 rows, 37 columns	Source records
Dashboard page	HTML, CSS, JavaScript, Chart.js 4.4.1	Filters, charts, KPI tiles and summary table
Data in the page	Columns packed as base64 binary and read as typed arrays	Lets the browser filter 100,000 rows
Website	HTML, CSS, JavaScript, DM Sans and Space Grotesk fonts	Project showcase with anchored sections
Web application	Python and Flask, named on the website as the app.py entry point	Web layer of the project
Verification	Python, pandas, headless Chrome	Dataset and page checks in Section 6
Version control	Git and GitHub	Stores the project

4. Project Design

4.1 Problem Solution Fit

Each problem identified in the ideation phase has a matching feature in the solution.

Problem	Solution feature
200,000 rows are too large to read	Six KPI tiles and four summary charts
Hard to compare slices of the market	Nine filters that update every chart together
Unclear how many vehicles a filter covers	Status line, such as Showing 100,000 of 100,000 vehicles
Findings need a written explanation	Tableau story with the main findings
Tableau is needed to open the analysis	Tableau Public views and a dashboard page that open in any browser

Problem	Solution feature
Views are spread across files and tools	One project website that links every view

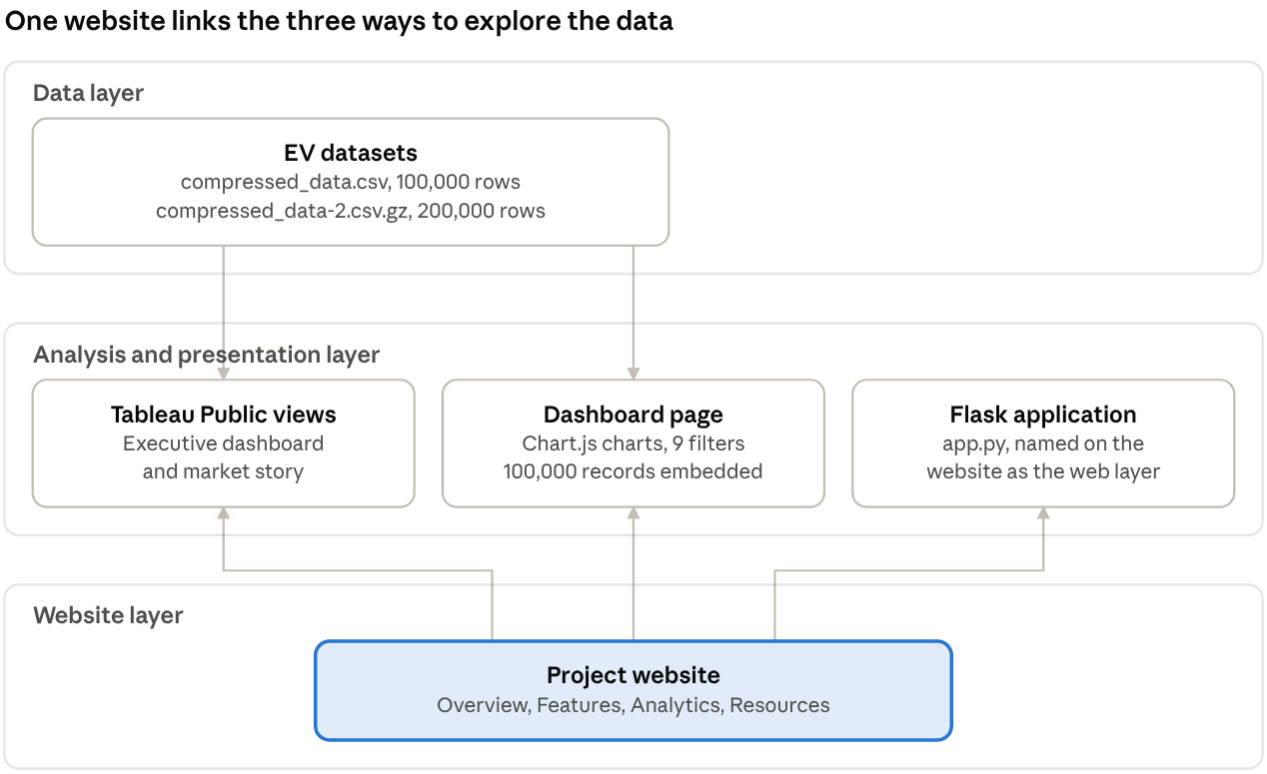
4.2 Proposed Solution

The proposed solution is a project website and a dashboard page, plus two Tableau Public views, all built on the same kind of EV data.

- **Project website.** A showcase page with Overview, Features, Analytics and Resources sections that links the repository, the report, the workbook and the live Tableau views.
- **Dashboard page.** Nine filters, six KPI tiles, four charts with a measure selector, a price-versus-range chart and a summary table, all drawn in the browser from 100,000 embedded records.
- **Tableau dashboard.** The executive dashboard, published on Tableau Public.
- **Tableau story.** The market story, published on Tableau Public.
- **Flask application.** The website names a Python Flask application, app.py, as the web layer of the project.

4.3 Solution Architecture

The architecture has a data layer, an analysis and presentation layer, and a website that links them.



architecture · 3 layers, 5 components

The website is the single entry point: it links the Tableau views, the dashboard page and the Flask application.

5. Project Planning & Scheduling

5.1 Project Planning

The project ran in five phases, from understanding the data to publishing the finished site.

Phase	Activities	Deliverable
1. Data understanding	Studied the 37-column EV dataset, a vehicle ID plus 36 attributes, and prepared two compressed files	compressed_data.csv with 100,000 rows and compressed_data-2.csv.gz with 200,000 rows
2. Analysis in Tableau	Built the executive dashboard and the market story and published both to Tableau Public	Live Tableau dashboard and story, with a workbook version dated 25 October 2026 in its address

Phase	Activities	Deliverable
3. Dashboard page	Packed 100,000 records into one HTML page with filters, charts and a summary table	EV Market Insights dashboard page
4. Project website	Built the showcase site with Overview, Features, Analytics and Resources sections	Project website, index-6.html
5. Documentation and release	Uploaded the files to GitHub and wrote this project report	Repository MahiPatidar8/EV-Market-, uploaded on 1 November 2026

6. Functional and Performance Testing

Twelve checks were run on the repository's own files: 9 pass, 2 pass with a data-quality note, and 1 fails when the dashboard page has no internet access. The dashboard page loads Chart.js from a CDN.

6.1 Performance Testing

Functional tests. The checks were run on 2 November 2026 with Python and pandas for the datasets and headless Google Chrome for the two pages, opened from disk.

Test	What it checks	Result
Dataset files load	Both files read: 100,000 and 200,000 rows with the same 37-column header	Pass
Unique vehicle IDs	The 100,000-row file has no repeated ID, the 200,000-row file has 1, and the two files share none	Note
Value ranges	Model years 2018 to 2026, ratings 1.52 to 5.0, resale value 15% to 90%, lowest price $900, shortest range 55 km	Pass
Blank values	18,996 blank cells in 5 columns of the 200,000-row file; the dashboard skips blanks when it averages	Note
Filters narrow results	Europe with NMC batteries gives fewer sales than Europe, which gives fewer than all regions	Pass
Multi-select adds up	Sales for 2020 and 2021 together equal the two years separately	Pass

Test	What it checks	Result
Empty selection	A filter with no matching vehicles returns 0 rows, and the page holds a no-match message	Pass
Website sections	Navigation links to Home, Overview, Features, Analytics and Resources each reach a section	Pass
Website on a phone	At 390 px wide the page cannot be scrolled sideways and shows a menu button	Pass
Website script errors	No script errors on a 1920 px load or a 390 px load	Pass
Dashboard filters	All 9 filters are filled with their options, such as 15 categories and 200 manufacturers	Pass
Dashboard charts offline	With cdn.jsdelivr.net unreachable, the charts and KPI tiles stay empty with the error Chart is not defined	Fail

Performance measurements. The figures below were measured on one run in the analysis environment, so absolute times will differ by machine.

Operation	Time
Read the 100,000-row CSV with pandas	1.60 s
Read the 200,000-row gzip CSV with pandas	1.30 s
Four dashboard-style aggregations, 100,000 rows	23.2 ms
Four dashboard-style aggregations, 200,000 rows	44.3 ms
Region and battery filter with a sales total, 200,000 rows	31.6 ms
Project website ready, desktop and phone width	0.57 s and 0.35 s
Dashboard page file of 7.5 MB loaded, before charts	0.63 s

The dashboard page needs no data server. It decodes its 100,000 embedded records once and filters them in the browser with one pass over the records on each change.

7. Results

The analysis shows a market dominated by small vehicles and two battery chemistries, with almost no difference between regions or years. The figures below are computed from the 200,000-record dataset.

- **Market size:** 1.55 billion annual sales units across 200 manufacturers and 17,980 models, with an average customer rating of 4.09 out of 5.
- **Vehicle categories:** Two-wheeler scooters lead with 174.6 million units, 11.3% of the total. Electric trucks, pickup trucks and buses each sell under 13 million.
- **Battery types:** NMC (524.1 million) and LFP (496.1 million) together hold about 66% of units. Sodium-ion has 153.3 million and solid-state 110.0 million.
- **Regions:** Totals sit between 217.5 million (Europe) and 224.0 million (South Asia), a gap of about 3%.
- **Range:** Luxury sedans average 550.6 km and luxury SUVs 524.8 km, while scooters average 105.0 km. Range and rating barely change by battery type: 324.1 to 325.9 km and 4.08 to 4.10.
- **Durability:** Solid-state batteries last about 2,997 cycles and lose 7.4% of capacity in five years. The other four types last 1,603 to 2,502 cycles and lose 10.1% to 14.0%.
- **Manufacturers:** The largest, AmpereWorks, sells 9.5 million units, about 0.6% of the market, so no manufacturer dominates.

7.1 Output Screenshots

Figure 1. Project website, home page. The hero section introduces the Flask web experience, Tableau analytics and data-driven insights.

Figure 2. Project website, Analytics section. Three entry points open the Flask website, the interactive Tableau dashboard and the Tableau story.

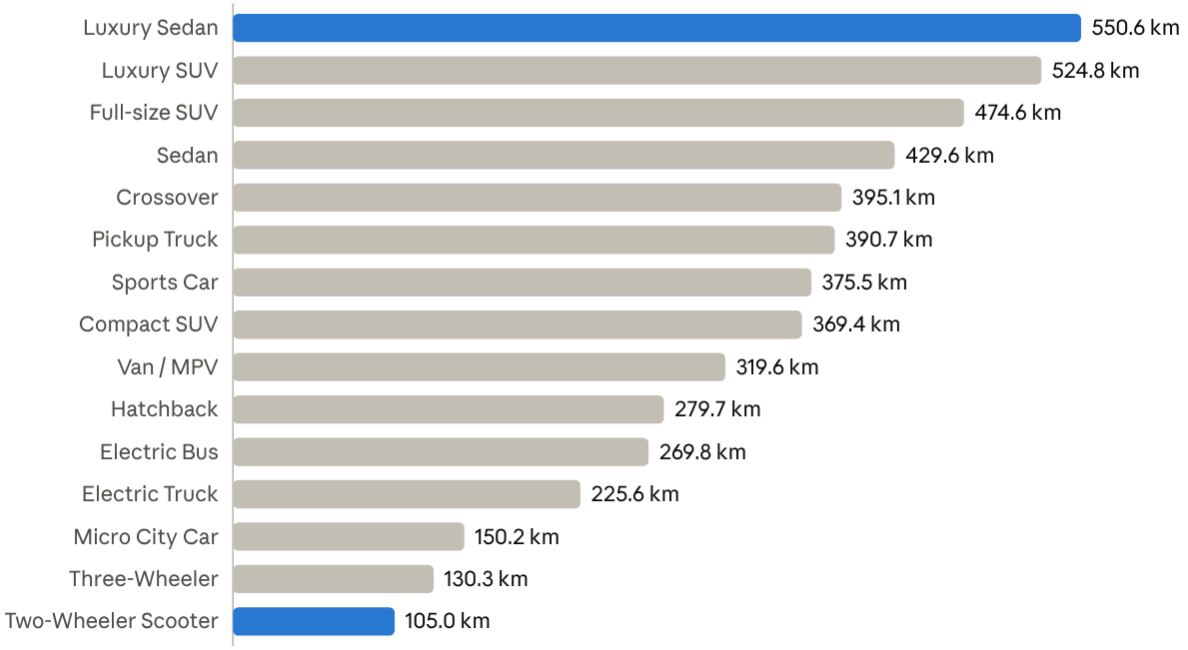
Figure 3. Units sold by vehicle category. Small vehicles sell the most and heavy vehicles the least.

compressed_data-2.csv.gz · 200,000 records, summed by vehicle category

Figure 4. Average range by vehicle category. Luxury sedans and SUVs travel farthest and scooters the shortest distance.

Luxury sedans average 550.6 km of range, over five times a scooter at 105.0 km

Average range in km on a full charge, by vehicle category



compressed_data-2.csv.gz · 200,000 records, averaged by vehicle category

8. Advantages & Disadvantages

The project makes a large dataset easy to explore, but its conclusions are limited by the data and by how the dashboard page loads its charts.

Advantages	Disadvantages
Every chart on the dashboard page updates together from one filter change	The data looks generated, so the findings are not real market facts
The dashboard page is one file with its data embedded, so it needs no data server	The dashboard charts need internet access, because Chart.js loads from a CDN
Tableau Public views open in any browser with no Tableau installation	The dashboard page uses the 100,000-row file, not the 200,000-row dataset analysed in this report
Nine filters, 12 measures and a copyable summary table give many views of the same data	18,996 blank cells in the larger file mean some averages use fewer records
One website links the repository, report, workbook and live views	The dashboard shows past data only, with no forecasting

9. Conclusion

EV Market Insights turns EV records into a project website, a filterable dashboard page and Tableau Public views that answer common market questions in seconds. The analysis shows a market led by two-wheeler scooters and by NMC and LFP batteries, with similar totals across regions and no dominant manufacturer. Checks on the repository's files show that the datasets, filters and website behave as expected, while the dashboard charts need internet access to load. Because the datasets appear to be generated, the project is best read as a working demonstration of an end-to-end analytics workflow, from Tableau analysis to a published website.

10. Future Scope

The project can grow in four directions, chiefly by replacing the generated data with real registration data.

- **Real data.** Load actual EV registrations and sales from official sources, so the findings describe the real market.
- **More attributes.** Use more of the 36 attributes, such as torque, top speed, insurance cost and warranty, to compare performance and ownership cost.
- **Forecasting.** Add trend lines and models that project sales by region and vehicle category.
- **Wider reach.** Bundle Chart.js with the dashboard page so it works offline, add data export and saved views, and host the Flask application on a cloud platform.

11. Appendix

Source Code

The repository holds the project website, the dashboard page and both datasets.

Item	Contents
index-6.html	Project website with Overview, Features, Analytics and Resources sections and two embedded Tableau Public views
EV Market Insights-2.html	Dashboard page with nine filters, six KPI tiles, four measure charts, a price vs range chart, a summary table and 100,000 embedded records
compressed_data.csv	100,000 EV records with 37 columns

Item	Contents
compressed_data-2.csv.gz	200,000 EV records with 37 columns, gzip-compressed
Live Tableau Dashboard, Live Tableau Story	Placeholder files for the two Tableau Public views

The website also links app.py, the Tableau workbook EV_Market_Performance_Market_Story.twbx and the report EV_Market_Performance_Final_Report-3.docx on GitHub.

Dataset Link

The datasets are stored in the GitHub repository: compressed_data.csv with 100,000 rows and compressed_data-2.csv.gz with 200,000 rows, each with 37 columns, a vehicle ID plus 36 attributes.

GitHub & Project Demo Link

- GitHub: github.com/MahiPatidar8/EV-Market-
- Tableau dashboard: EV Executive Dashboard on Tableau Public
- Tableau story: Market story on Tableau Public
- Project demo: add the hosted link here once the site is published.